

Quantitative relationship between resting heart rate reduction and magnitude of clinical benefits in post-myocardial infarction: a meta-regression of randomized clinical trials

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Received 12 June 2007; revised 13 September 2007; accepted 26 September 2007; online publish-ahead-of-print 2 November 2007

KEYWORDS

Meta-regression;
Resting heart rate reduction;
Beta-blockers;
Calcium channel blockers;
Meta-analysis;
Mortality

Aims The impact on mortality outcomes of beta-blockers and calcium blockers in post-myocardial infarction (MI) has been suggested to be related to resting heart rate (HR) reduction. A meta-regression of randomized clinical trials was carried out to assess this relationship using weighted meta-regression of logarithm of odds ratio against absolute HR reduction.

Methods and results Twenty-five controlled randomized trials (21 with beta-blockers and four with calcium channel blockers) involving a total of 30 904 patients meet eligibility criteria, but only 17 documented changes in resting HR (14 with beta-blockers and three with calcium channel blockers).

A statistically significant relationship was found between resting HR reduction and the clinical benefit including reduction in cardiac death ($P < 0.001$), all-cause death ($P = 0.008$), sudden death ($P = 0.015$), and non-fatal MI recurrence ($P = 0.024$). Each 10 b.p.m. reduction in the HR is estimated to reduce the relative risk of cardiac death by 30%.

Conclusion The meta-regression of the randomized clinical trials strongly suggest that the beneficial effect of beta-blockers and calcium channel blockers in post-MI patients is proportionally related to resting HR reduction. Furthermore, the absence of residual heterogeneity indicated that resting HR reduction could be a major determinant of the clinical benefit.

Introduction

Epidemiological studies suggest that lower resting heart rate (HR) is associated with decreased cardiovascular and all-cause mortality.^{1–9} HR has also been reported to be an independent predictor of bad outcome after myocardial infarction (MI)^{10,11} and a determinant of infarct size in acute MI.^{12,13}

A relationship between the resting HR reduction and the mortality has also been observed in post-MI patients with beta-blockers. In the Norwegian timolol study,¹³ logistic regression analysis of the total death relative to the HR at 1 month after the start of treatment (with timolol or placebo) remove the difference in outcome observed between the timolol group and the placebo group. On the basis of the results of 10 randomized controlled trials, Kjekshus¹⁴ had observed that a correlation may exist between beta-blocker-induced reduction in resting HR and reduction in total mortality. So a growing body of evidence also suggests that pharmacological reduction in resting HR could decrease morbidity and mortality in cardiovascular patients.¹⁰

We sought to address this issue by meta-regression. Meta-regression aims to investigate whether particular covariates (potential effect modifier) explain any of the difference in treatment effects between multiple studies.¹⁵ A meta-regression was performed to determine to which extent resting HR reduction induced by the various drugs modifying HR affects the reduction of mortality and morbidity observed in randomized placebo-controlled trials with these drugs in post-MI.

Methods

The meta-regression was performed according to pre-defined selection criteria for trial search and data analysis. QUOROM standards were followed¹⁶ during all phases of the design and implementation of this meta-regression.

Study identification

I identified relevant published and unpublished unconfounded randomized trials that compared beta-blockers and calcium blockers with placebo in post MI. I searched electronic databases (PubMed, Embase) from 1966 to 1 January 2006 and the Cochrane Controlled Trials Register (CENTRAL Issue 4, 2005). The broad strategy

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described by Haynes¹⁷ was used for PubMed (search strategies are described in the Supplementary material online).

I also reviewed the bibliographic references of the retrieved studies, review and meta-analysis articles obtained from the initial search. Conference proceedings of the following conferences were hand searched: AHA, ACC, ESC from 1995 to 2005.

I also searched the WEB with the same keywords and trial registers (www.clinicaltrialresults.org and ISRCTN register).

I included trials published only in abstract form, to limit the influence of possible publication bias.

Study selection

I assessed all potentially relevant published articles and abstracts for inclusion. To be included, trials had to meet the following criteria: (i) randomized placebo controlled (with or without allocation concealment); (ii) double-blind design; (iii) with <10% attrition; (iv) patient followed-up for 1 year or more. Trials initially planned for 1 year follow-up or more but stopped prematurely for efficacy reasons were also considered; (v) patients with a history of MI regardless of the period of treatment initiation. Trials with intravenous treatment initiated during the acute phase of MI were also included; (vi) trials had to be conducted with a beta-blocker, including those with partial beta-1 agonist activity or with a calcium channel blocker.

Data collection and assessment of quality

All qualifying trials were assessed for adequate blinding of randomization, completeness of follow-up, and description of withdrawals using Jadad score.¹⁸

The absolute HR reduction was calculated by subtracting the mean change from baseline observed in the placebo group from the corresponding change observed in the active treatment group. Changes in resting HR were determined between baseline resting HR and resting HR after about 1 month on treatment, the exact time of measure varying across trials. One month is the best compromise between the stabilization of the HR reduction and the number of patients at risk and under treatment. Over this time, occurrence of death and withdrawals decreased the number of patients contributing towards the HR mean.

Statistical methods

An initial robust analysis was performed to search if a relationship between resting HR reduction and clinical benefit exists. Clinical benefit refers to the estimation of the treatment effect on the clinically relevant endpoints of mortality and morbidity. Odds ratio between the active treatment and placebo group was used to estimate the clinical benefit on these endpoints. The trials were split in three subgroups according to tertiles of HR reduction. For each variable of clinical benefit, a pooled odds ratio was calculated in every subgroup of resting HR reduction using a fixed model, given an estimate of the effect of treatment for the higher, average and lower resting HR reductions. Then the three estimates were compared with the χ^2 test for trend.¹⁹

In the second step, the analysis used a weighted meta-regression by the inverse of variance,^{15,20} modelling the logarithm of odds ratio as a linear function of the absolute resting HR reduction. In this meta-regression, we used additive component of residual heterogeneity in order to take into account the diversity between trials regarding drugs, regimen and patients. Restricted maximum likelihood (REML) estimators were used.²¹ See online supplementary data for model details.

Slope estimates were used to predict the relative risk reduction potentially induced by 10 b.p.m. reduction in resting HR.

The robustness of the relationship was tested by sensitivity analyses where REML estimates were computed after exclusion of trials with the lowest and largest resting HR reduction (one each time). Subgroup analyses were planned to explore separately beta-blockers and calcium blockers.

As a complementary analysis (planned in the protocol), we performed also the meta-regression using the relative risk. Publication bias was assessed graphically using a funnel plot of the logarithm of effect size vs. the standard error for each trial.

The meta-regression methods were implemented with the R software.²² The code was validated with running examples reported in the methodological papers.

Statistical test were two-tailed with $P < 0.05$ chosen at the level of significance.

Results

Study selection

The process of study selection is outlined in *Figures 1* and *2*. After selection, 21 trials with beta-blockers were included and nine excluded, whereas four trials with calcium channel blockers were included and 15 excluded. The more frequent reason for exclusion was an insufficient follow-up duration of <1 year. (The lists of excluded trials are given in the Supplementary material online.) Resting HR reduction was reported in 14 of the 21 trials with beta-blockers and in three of the four trials with calcium channel blockers giving a total of 17 studies applicable for the meta-regression.

The data concerning the HR reduction being not available for all trials, we undertook a subgroup meta-analysis to compare trials reporting HR reduction to the others in

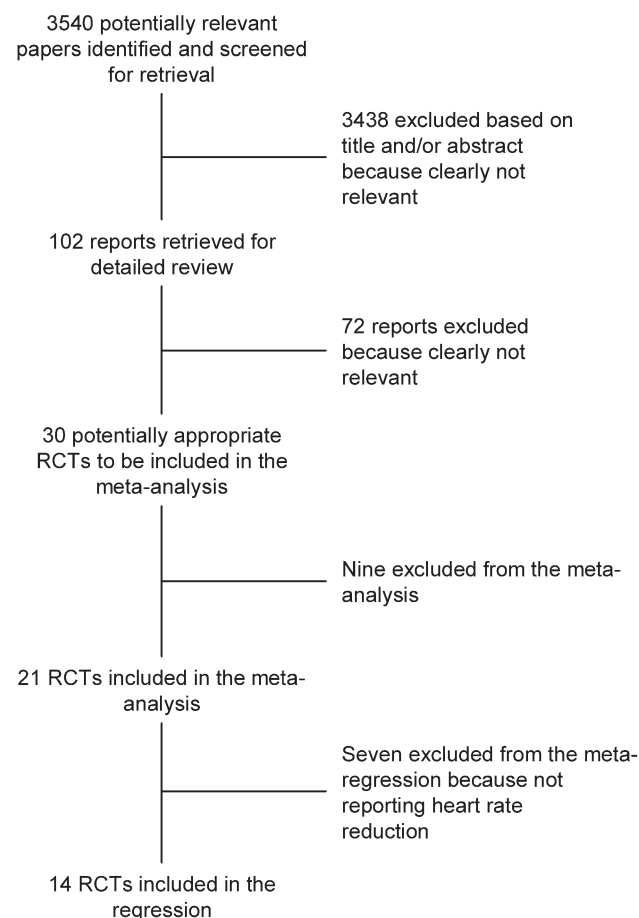


Figure 1 Process of beta-blockers trials selection.

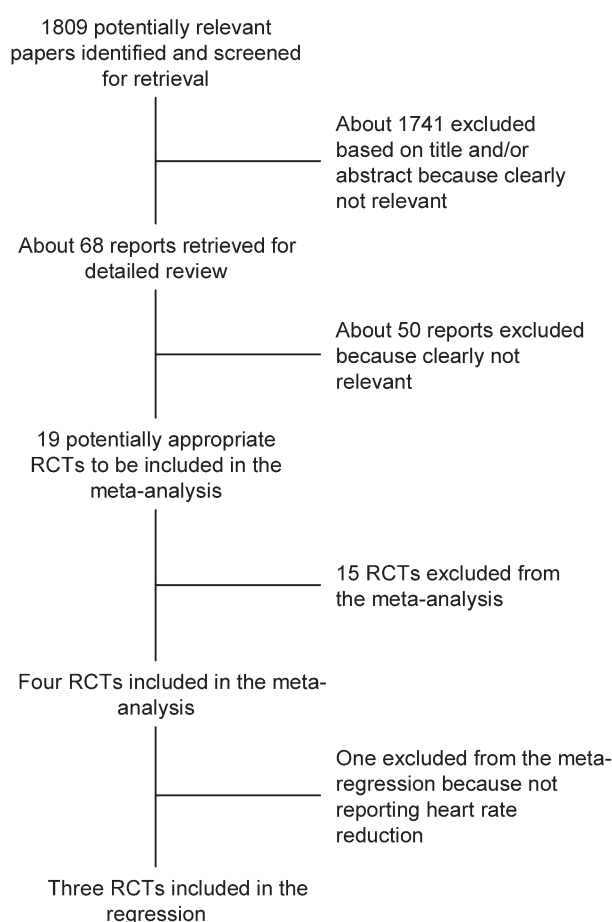


Figure 2 Process of calcium channel blockers trials selection.

terms of treatment effects using the relative risk and a fixed effect model. No statistically significant heterogeneity was found between trials reporting the HR reduction value and those not reporting this data.

Included trials

Beta-blockers and calcium channel blockers used in analysed trials are shown in *Table 1*.

Inclusion criteria of all the trials specified that enrolled patients had definite or suspected MI. Depending of the protocol of the trial, MI was diagnosed using clinical, electrical signs and enzyme elevation (alone or in combination). The enzymatic definition of MI varied across time and therefore across the trials.

Beta-blockers trials were quite well conducted with random allocation of treatment, masking of treatment assignment, and patient follow-up rates of >95% except for one trial (Wilhelmsson 1974 in which 7% of patients were lost to follow-up). However, six trials had a Jadad score less than 4, mainly due to insufficient description of randomization or double-blind, preventing the evaluation of their appropriateness (allocation concealment in particular) or due to absence of withdrawals description. The calcium channel blockers trials were also well conducted and obtained all a Jadad score of 4 or 5.

Effects of resting heart rate reduction on clinical benefit

According to the availability of resting HR reduction and endpoints data, 16 trials contributed to the meta-regression for all-cause death, 12 for cardiac death, 7 for cardiovascular death, 13 for non-fatal MI recurrence and 6 for sudden death.

Subgroups analysis by resting HR reduction tertiles (*Table 2* and *Figures 3–6*) showed that larger resting HR reduction, compared with lower resting HR reduction, were associated with an increased mortality reduction for cardiac death ($P = 0.0015$), all-cause death ($P = 0.017$) and sudden death ($P = 0.005$). Similar relationship was found for non-fatal MI recurrence ($P = 0.033$). No significant relationship was found between resting HR reduction and the treatment effect on cardiovascular deaths ($P = 0.09$) and on fatal and non-fatal MI ($P = 0.36$). Cardiac events were available in only three trials and did not allow any reliable estimate of the relationship.

These results were all confirmed by the meta-regression (*Table 2* and *Figures 3–6*) which showed the same type of relationship.

No residual heterogeneity was detected, except for the cardiovascular death where a marginal heterogeneity appeared.

In the sensitivity analysis, the results for cardiac death remained statistically significant after the exclusion of trials with highest or lowest resting HR reduction. For all-cause mortality, sudden death, and recurrence of MI, results remain significant after removing the trial with the smallest resting HR reduction. Results on the same endpoints became statistically non-significant after removing the trial with the largest resting HR reduction but the results overall remain qualitatively unchanged.

The meta-regression using log-relative risk (in place of log odds ratio) led to very similar results (*Table 3*).

When analysis is restricted to the beta-blockers trials, a statistically significant relationship was found between HR reduction and log odds ratio for cardiac death ($P = 0.02$, meta-regression slope=0.039), sudden death ($P < 0.01$) and non-fatal MI recurrence ($P < 0.01$). A similar but non-statistically significant relationship ($P = 0.17$, meta-regression slope=0.21) was found with all-cause mortality. Given the small number of available data (3 points), meta-regression restricted to the calcium blocker trials was not performed.

Funnel plots did not suggest the possibility of publication bias.

Discussion

The present meta-regression shows a relationship between the clinical benefit and the resting HR reduction observed with drugs modifying HR in post-MI patients. These findings provide firm evidence that the clinical benefit on cardiac death, all-cause mortality, sudden death and non-fatal recurrence of MI is proportional to the extent of resting HR reduction.

For all these endpoints, the residual heterogeneity is estimated at zero, showing that the differences between trials can be completely explained by resting HR reduction and within-trial variability (sampling fluctuations). This result is

Table 1 Characteristics of the trials included in the meta-regression

Trial	Studied treatment	HR at baseline	Absolute HR change	Age, mean (years)	Female (%)	n (treated/control)	Follow-up duration (mean)	Attrition, n (%)	Inclusion period	Jadad's score
Beta-blockers										
B1 Andersen <i>et al.</i> ³⁰	Alprenolol	NA	NA	NA	NA	238/242	About 1 year	0	March 1976–December 1978	3
B2 APSI ³¹	Acebutolol	82.8/81.8	−8.8	NA	27	298/309	318 days	0	April 1987–September 1988	5
B3 Aronow <i>et al.</i> ³²	Propranolol	NA	NA	81	70	79/79	1 year	Unclear	NA	2
B4 Australian and Swedish Study ³³	Pindolol	77.9/76.8	−5	58	17	263/266	2 years	Unclear	February 1978–January 1980	4
B5 Baber <i>et al.</i> ³⁴	Propranolol	81.3/81.9	−13	55	15	355/365	9 months	Unclear	NA	1
B6 Basu <i>et al.</i> ³⁵	Carvedilol	NA	−13.5	60	23	77/74	6 months	5 (3.31%)	February 1992–September 1994	5
B7 BHAT ^{36–38}	Propranolol	76.2/75.7	−8	55	36	1916/1921	25 months	12 (0.3%)	June 1978–October 1980	5
B8 EIS ^{39,40}	Oxprenolol	73.5/74.2	−8	55	29	858/883	1 year	Unclear	July 1979–July 1981	4
B9 Hansteen <i>et al.</i> ¹⁹	Propranolol	81.5/78.7	−14	NA	31	278/282	1 year	0	December 1977–July 1980	5
B10 Hjalmarson <i>et al.</i> ⁴¹	Metoprolol	NA	−13	40–74 (range)	35	698/697	2 years ^a	1.60%	June 1981–January 1981	5
B11 Julian <i>et al.</i> ⁴²	Sotalol	76.4/77.3	−15.8	55	26	873/583	12 months	0	January 1978–August 1980	4
B12 LIT Research Group ⁴³	Metoprolol	79/79	NA	58	25	1195/1200	18 months	0.20%	August 1979–April 1982	5
B13 Manger Cats <i>et al.</i> ⁴⁴	Metoprolol	NA	NA	NA	NA	273/280	1 year	NA	NA	3
B14 Multicentre international ⁴⁵	Practolol	76.8/77.2	−9	55	36	1533/1520	12 months, up to 24 months	3.40%	NA	5
B15 Norwegian Multicentre Study Group ⁴⁶	Timolol	74.4/74.3	−18	61	38	945/939	17 months	Unclear	January 1978–October 1979	4
B16 Rehnqvist <i>et al.</i> ⁴⁷	Metoprolol	NA	NA	NA	NA	154/147	36 months	NA	NA	2
B17 Salathia <i>et al.</i> ⁴⁸	Metoprolol	NA	NA	NA	46	416/384	1 year	0.50%	NA	5
B18 Schwartz <i>et al.</i> (high risk and low risk) ⁴⁹	Oxprenolol	NA	NA	NA	NA	485/488	22 months (at least 6 months)	NA	NA	4
B19 Taylor <i>et al.</i> ⁵⁰	Oxprenolol	76/77	−6	51	NA	632/471	48 months	Unclear	1973	4
B20 Wilcox <i>et al.</i> ⁵¹	Propranolol	79.5/80	−17	NA	28	259/129	1 year	0	NA	5
B21 Wilhelmsson <i>et al.</i> ⁵²	Alprenolol	NA	−6	NA	NA	114/116	2 years	(16) 7%	January 68–	3
Calcium blockers										
C1 MDPIT <i>et al.</i> ⁵³	Diltiazem	71/72	−3	48	20	1234/1232	25 months (at least 12 months)	3.20%	February 1983–June 1986	5
C2 CRIS ⁵⁴	Verapamil SR	73/74	−2	55	9	531/542	23.5 months	5 (0.47%)	1985–87	5
C3 DAVIT II ⁵⁵	Verapamil	75/75	−6	NA	20	878/897	16 months	NA	February 1985–	4
C4 SPRINT I ⁵⁶	Nifedipine	76.9/77.4	NA	58	15	1130/1146	1 year	NA	August 1981–July 1983	4

^aIn the Göteborg Metoprolol Study (B10), the double blind and placebo-controlled study lasted for 3 months. After that all patients were given open treatment with metoprolol and outcome was reported after 2 years.

Table 2 Result summary of the analysis by subgroup and meta-regression

Endpoint	Number of trials with sufficient data	Subgroup analysis by HR reduction tertile, <i>P</i> -value for trend	Meta-regression slope	Meta-regression, <i>P</i> -value for slope	Residual heterogeneity	Per cent reduction in odds ratio for 10 and 15 b.p.m. reduction in resting HR ^a
All-cause death	16	0.017	-0.0249	0.008 ^b	0	22/31
Cardiac death	12	0.002	-0.0396	<0.001 ^c	0	33/45
Cardiovascular death ^d	7	0.090	-0.0204	0.550	0.0891	—
Sudden death	6	0.005	-0.0531	0.015 ^b	0	41/55
Non-fatal MI recurrence	12	0.033	-0.0243	0.024 ^b	0	22/31
Fatal and non-fatal MI	6	0.360	0.00227	0.960	—	—
Cardiac event (fatal and non-fatal)	3	—	—	—	—	—

^aPer cent reduction in odds ratio are obtained by the exponential of the meta-regression slope.

^bRobustness result remains statistically significant only after removal of the lowest HR reduction. The removal of the largest HR implies a NS result.

^cRobustness result remains statistically significant after removal of the trial with the largest HR reduction or those with the smallest.

^dCardiovascular deaths add to cardiac deaths vascular deaths and strokes.

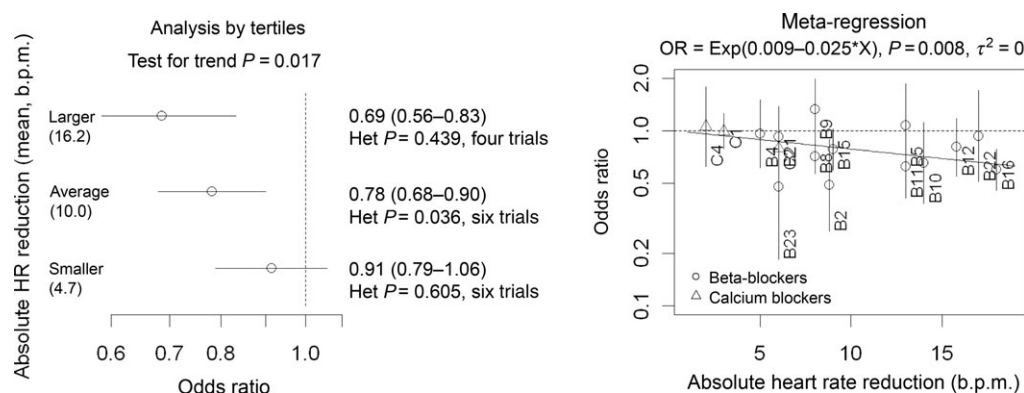


Figure 3 By tertiles subgroup analysis and meta-regression for all-cause death. Odds ratios represented are comparing odds between the active treatment and placebo group.

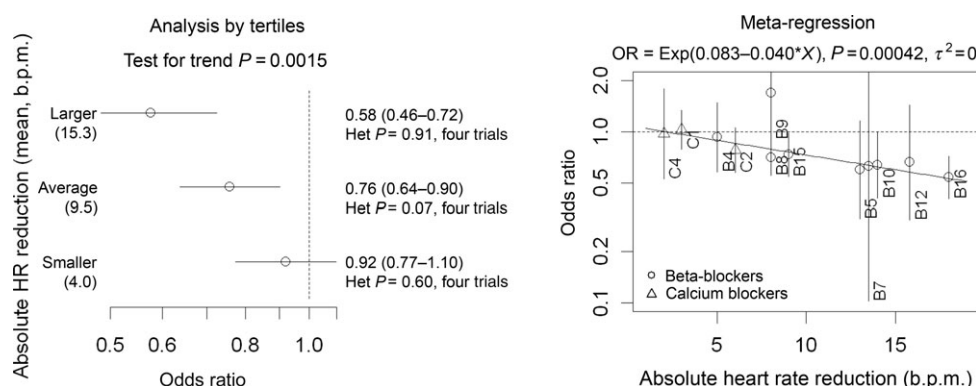


Figure 4 By tertiles subgroup analysis and meta-regression for cardiac death. Odds ratios represented are comparing odds between the active treatment and placebo group.

compatible with the assumption that the resting HR reduction explains all the benefit of these drugs. From this meta-regression, there is no evidence of a drug/class specific part in the benefit. The lack of mortality reduction observed with calcium antagonists could be totally explained by the absence of resting HR reduction with the tested drugs without need to take into account a specificity of the calcium blockers. After adjustment on resting HR reduction,

there is no residual heterogeneity to be explained by other factors between the trials. Thus, resting HR reduction appears to be the major determinant of the clinical benefit induced by the drugs modifying resting HR in post MI patients.

For cardiovascular deaths, no statistically significant relationship was found most probably due to the small number of trials available for this endpoint, seven in place of 12 for cardiac deaths. The value of the slope for this endpoint

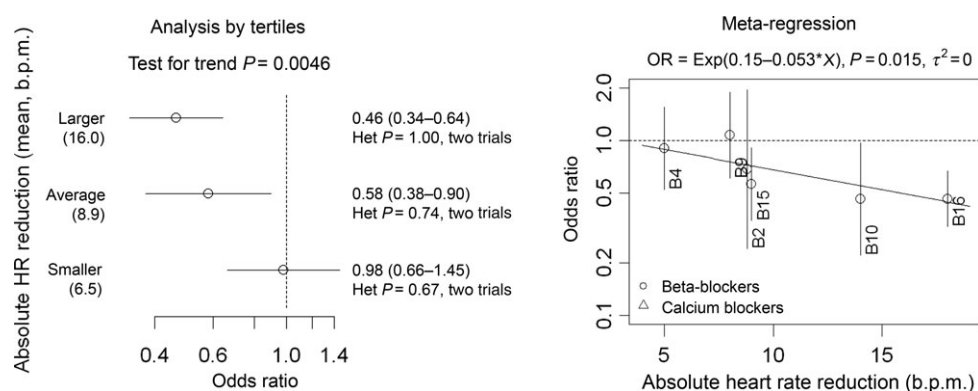


Figure 5 By tertiles subgroup analysis and meta-regression for sudden death. Odds ratios represented are comparing odds between the active treatment and placebo group. There were no data available for sudden death in the calcium blockers trials.

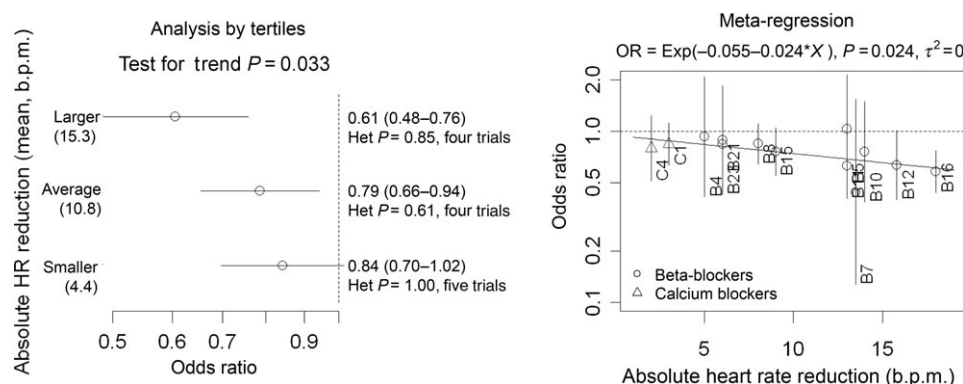


Figure 6 By tertiles subgroup analysis and meta-regression for non-fatal myocardial infarction recurrence. Odds ratios represented are comparing odds between the active treatment and placebo group.

Table 3 Comparison of the results given by the meta-regression with the odds ratio and the relative risk

Endpoint	Subgroup analysis by HR reduction tertile, P -value for trend		Meta-regression, P -value for slope		Per cent reduction for 10 b.p.m. reduction in HR	
	Odds ratio	Relative risk	Odds ratio	Relative risk	Odds ratio	Relative risk
All-cause death	0.017	0.015	0.008	0.008	21	20
Cardiac death	0.002	0.002	<0.001	<0.001	33	30
Cardiovascular death	0.090	0.110	0.550	0.570	—	—
Sudden death	0.005	0.005	0.015	0.016	41	39
Non-fatal MI recurrence	0.033	0.030	0.024	0.020	22	21
Fatal and non-fatal MI	0.360	0.320	0.960	0.990	—	—
Cardiac event (fatal and non-fatal)	—	—	—	—	—	—

–0.0204 is similar to the one observed with the cardiac deaths –0.0396. Moreover, the relationship could have been weakened by the adding of the vascular deaths (including among others fatal stroke) not influenced by the HR.

The meta-regression was undertaken using the odds ratio for statistical reasons. The odds ratio has symmetric properties that the relative risk does not have; therefore, odds ratio is more appropriate to use for a meta-regression than the relative risk. However, in practice, it appears that the relative risk is more relevant for clinicians (among others)

than odds ratio.^{23,24} The analyses carried out on relative risk gave consistent results with analyses performed on odds ratio. For the sake of simplicity, the relative risk can be used to report the findings for this meta-regression.

Our results are consistent with those obtained by Kjekshus^{14,25} in 1986 (updated in 1999) when considering only beta-blockers. Our results are not simply the replication of these previous ones, given that five additional beta-blocker trials were added and that the focus was widened to the other drugs modifying HR.

In previous meta-analyses, only the intrinsic sympathomimetic activity has been identified as a potential effect modifier of the beta-blockers benefit^{26,27} and could be, thereby, a confounding factor for the relation between resting HR reduction and benefit. However, a main consequence of the intrinsic sympathomimetic activity is a low reduction in resting HR. In fact, the relation between resting HR reduction and benefit could explain the trend towards a decreased benefit with drugs with intrinsic sympathomimetic activity.

Some potential limitations must be discussed. Data dredging is the main pitfall in reaching reliable conclusion from meta-regression.²⁸ Here, it can be discarded because the covariate was pre-specified. Clearly, this work is hypothesis testing and not exploratory like an intensive search among a large collection of candidate covariables would be.

These results are quite robust. The relationship is found with robust subgroup analysis that did not require strong statistical assumptions. Moreover, the validity of these observations is strengthened by the robustness of the sensitivity analyses and by the very good fitting with the absence of residual heterogeneity.

The relationship with resting HR lowering may be potentially confounded by other trials, drugs, or patients characteristics. Given the relatively small number of trials reporting resting HR reduction, multivariate analyses are not feasible in this analysis based on summary data for each considered trials. Blood pressure reduction is a candidate confounding factor as it is probable that blood pressure lowering induced by these drugs is in part correlated with the induced resting HR reduction. Despite that statistical adjustment was not feasible, it is improbable that the found relationships were confounded by blood pressure change. In randomized trials, clinical benefit of blood pressure lowering needed several years to appear and mortality reductions were smaller than those observed in the present post-MI trials.²⁹ Moreover, in the trials used in this meta-regression, patients were included irrespective to the presence of a hypertension.

The between trial heterogeneity concerning drugs and patients is taken into account with the use of a random effect model. The absence of heterogeneity does not resolve by itself the confounding concern but gives some reassurance that the extent of resting HR lowering is a strong marker of clinical benefit in the trials.

An interesting result of this meta-regression is the absence of residual heterogeneity that could be interpreted as all the benefit is brought by resting HR reduction without any other mechanism. This attractive interpretation is limited as it derives from non-statistically significant results. However, it is at least reasonable to conclude that the available data do not permit to exclude that the HR reduction is the major determinant of the benefit of drugs like beta-blockers or calcium blockers in post-MI.

In conclusion, in post-MI condition, this meta-regression of randomized clinical trials robustly suggests that the benefit of drugs modifying HR is strongly related to the magnitude of reduction in resting HR. This implies that whatever the mechanism leading to the decrease in resting HR, a same reduction could result in the same morbidity and mortality reduction. Each 10 b.p.m. reduction in resting HR is estimated to reduce the relative risk of cardiac death by about 30%, the risk of sudden death by 39%, leading to a reduction in the relative risk of all-cause mortality of 20%.

Supplementary material

Supplementary material is available at European Heart Journal online.

Acknowledgement

We thank D. Lebrasseur, I.R.I.S and the scientific documentation department of I.R.I.S. for their help in bibliographic searching.

Conflict of interest: M.C. has received research support from and is a consultant to Servier.

Funding

This study was supported by an unrestricted grant from Servier, Inc. The sponsor of the study had no role in study design, data collection, data analysis, data interpretation, or in the writing of the report.

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